

We thank the AE and reviewers for reading our manuscript and for the thoughtful comments and suggestions which have improved the clarity and quality of the paper. We have attempted to adequately address all of the concerns. Point-by-point responses are included below and all modified text has been highlighted in red in our revised submission.

AE

The reviewers agree that the manuscript addresses an important and timely problem and view the modeling framework as technically strong and thoughtfully developed. They are generally positive about the simulation study and the real-data applications. At the same time, the reviews emphasize the need for clearer motivation and interpretation of the modeling choices, particularly around the added complexity of the SIHRD structure, the role of undetected infections, and the behavioral alarm function. They also request a clearer explanation of assumptions, implementation details, and how to interpret results, especially when comparing model variants. Please address the reviewers' comments carefully and systematically.

Additional comments

1. Beyond addressing the specific points raised, I encourage you to sharpen the narrative arc of the paper: clearly articulate early on what new scientific or policy-relevant insights this framework enables that simpler behavioral or compartmental models cannot, and return to those insights explicitly in the discussion.

✓ Thank you for this comment. We have significantly overhauled the introduction to our paper to sharpen the narrative arc and articulate the new scientific insights that our framework enables.

2. You may also consider modestly streamlining the presentation by moving some secondary methodological detail to supplementary material, which would help keep the main text focused on the key modeling contributions and substantive conclusions.

Thank you for this suggestion. We have moved some of the lengthier parts of the discussion to the supplementary material and also some of our revisions to the introduction have removed some extra technical details and streamlined the presentation.

Commented [CW1]: To Rob and Alex, I did not move anything in the methods, simulation study, or data analysis. This reads to me as a suggestion and not requirement and it is not specific enough to be helpful.

I feel that everything in those sections contributes significantly to the paper, but I am biased. If there are any parts you think could be moved to the supplement please let me know.

I think §2.3 could go in the supplementary with just a brief reference to your paper. I'm not sure that would actually improve the flow, but it might please the AE.

on alternative alarm formulations

Reviewer 1

This manuscript describes a Bayesian stochastic Susceptible-Infectious-Hospitalized-Recovered-Dead model which incorporates undetected infections and allows the transmission rate to vary in response to the number of cases and deaths through an 'alarm' function which captures the impact of dynamic human behavior on transmission. The paper is well written and addresses an important modeling issue. The simulation study is well-designed and the data application is impactful. I have just a few suggestions for improving the paper.

Thank you for your positive feedback and thoughtful comments. They were very helpful for improving the paper.

Major comments

1. When the model assumes the presence of undetected infections, it seems reasonable that hospitalizations would only arise from detected infections (at least in the context of the COVID pandemic, where everyone hospitalized with respiratory systems would have been tested for COVID and the tests were reasonably accurate). However the assumed model allows hospitalizations to arise from any infection (I rather than C). Can the authors comment on the implications of this assumption? Would restricting the model to require infections to arise only from detected infections improve the identifiability of parameters involved in the transitions into the hospitalized state? This seems possible, as we have more direct information about the number of detected infections than the total number of infections.

This is an excellent point and assuming that everyone hospitalized arises from the case counts does in fact improve the identifiability of model parameters, specifically the parameters λ which governs the I to H transition and γ_1 (I to R transition rate). We have now included some supplementary simulations assessing this revised model structure in the Supplement (Section 4), and mentioned this in lines XX of the main text.

The reason for our choice of allowing hospitalizations arise from any infection was because while it's clear that everyone in the hospitalization data would have received a positive test, it's not clear whether they would still be part of the reported cases counts if that test occurred after hospitalization. As real-world data do not provide a link between individuals in the case counts and hospitalization, it's difficult to assess whether this assumption would hold in practice. Therefore, we thought it would be best to keep the modeling framework more general.

I would put it in the discussion I think.

Commented [CW2]: I haven't actually added anything to the main text yet but I will. Do you think it makes more sense to add in the methods section after introducing the SIHRD structure or in the discussion section?



2. Can the authors comment on why excluding the alarm function from the model results in infections being so severely underestimated? I would have assumed removing the alarm function would cause a higher transmission rate and therefore infections to be overestimated.

We have added remarks about this behavior in both the simulation results (line 310-311) and data analysis results (line 383). To summarize briefly: the models without an alarm function must estimate an average transmission rate across the entire time period being modeled. When this window includes periods of both increasing and decreasing case counts, the estimated reproductive number ends up being right about 1 which leads to posterior predictive distributions with much less cases than was observed.

many fewer were

3. Page 12, starting line 290. In the description of the data, it's unclear if the hospitalization data consisted of the number of new hospital admissions each day or the total number of hospitalized patients each data. Line 282 specifically says the number of 'admitted patients' for Miami while line 285 says 'counts' for Montreal. Were these different quantities? To me, the word 'admitted' implies a new patient while count implies a census.

Thank you for letting us know that this was not clear. In both cases the data provides new admissions which directly corresponds with H^* in our model. We have revised the language in this section (Lines 340 - 343) to be clearer.

4. The authors should briefly describe how the initial conditions (S_0 , I_0 , H_0 , R_0 , D_0) were specified (or what priors were used to estimate them) for the real data applications.

Thank you for noting that we had omitted any description of the initial conditions and the priors used. This is an important part of implementing these models in practice and we have revised the text around lines 358-360 to provide further details, with the full priors being specified in the supplementary material section 5.1.

5. In figure 6, it is unclear what timeframe of data was available to the models which made these predictions. Were the models trained on data from the entire time period shown on the x-axis or does a portion of these plots represent out of sample prediction? If the figure does not include out of sample prediction, an assessment of the model's out of sample prediction would be very helpful. If the figure does include out of sample prediction, some indication of which time portion corresponds to out of sample prediction would be helpful.

These are plots of posterior predictive fit, so all data shown was used for model fitting and then forecasting is based on repeated draws from the posterior as describe in Section 2.5. We have slightly revised the text in Section 2.5 to be clearer about this process.

Previous works (Ward et al. 2023, Ward et al. 2025) have evaluated out of sample posterior predictive forecasting for alarm-based behavioral change models in simulation settings and found that various alarm function specifications could lead to accurate forecasting in settings of behavioral change. We did not evaluate out of sample predictive performance in this work, as this has been assessed previously and our primary goal was to make more accurate inference about past behavioral change. We have clarified this purpose in our revisions to the introduction.

Ward C, Deardon R, Schmidt AM. Bayesian modeling of dynamic behavioral change during an epidemic. *Infectious Disease Modelling*. 2023;8(4):947–963.464

Ward, M.A., Deardon, R. and Deeth, L.E. (2025). A framework for incorporating behavioural change into individual-level spatial epidemic models. *Can J Statistics*, 53: e11828. <https://doi.org/10.1002/cjs.11828>

Minor comments

1. Page 2 line 40: Foot and Mouth Disease is generally not capitalized. **fixed**
2. Page 2 line 49: The 'V' in the SIHVR model of Self et al. stands for ventilated, not vaccinated. The model did not include vaccinations. **fixed**
3. Page 2 line 58: The 'chain' in Markov Chain Monte Carlo should not be capitalized. **fixed**
4. Page 4, line 75: It should be 'they have' rather than 'they've'. **fixed**
5. Page 4, line 78: It would be good to clarify that use of the IDD transmission probability means that R_t values are no longer binomially distributed as stated on page 3 line 74 **we have clarified this on lines 122-123**
6. Page 7 line 197: The grammar of the first sentence is not quite right, as the first part of the sentence refers to the subsequent clauses as 'objectives' but the clauses themselves lack verbs. This could be remedied by adding 'determine' (or similar word) to the opening of each objective. **fixed**
7. The caption of Figure 1 states that infections were not plotted on the 'high deaths importance' figure, but that pane does appear to show infections (grey lines?) **thank you for catching this. The figure was supposed to have a truncated y-axis to better visualize differences in the observed cases, hospitalizations, and deaths, and this has been updated so the caption is now accurate.**
8. Are the results displayed in figures 3 and 4 from the models that included or ignored undetected cases? **These results are restricted to models the included undetected cases.**

We have updated the figure captions and text (lines 292-294) to clarify this.

9. Page 16 line 359: I believe it should be 'illustrated' rather than illustrate. **fixed**

10. Page 16 line 373: It should be 'do not' rather than 'don't'. **fixed**

Reviewer 2

Overall, this work tackles an important area of epidemiology: how to incorporate behavior into epidemic models. The paper is clearly presented and technically rich. However, several conceptual and modeling choices need to be clarified. My comments are meant to help the authors better communicate their findings to the readers.

Thank you for your thorough feedback. We are grateful as this has greatly improved the motivation for our work and communication of our findings. We have made substantial revisions to the introduction and throughout to address your comments.

Major comment 1: Inadequate motivation of the study

In the introduction, the authors failed to provide enough motivation for the extensions proposed in this work. Improvements are needed to clearly articulate why this extension was pursued so that it does not read like they are adding unnecessary complexity.

Thank you for this important critique. We have substantially revised the introduction of the paper to clearly articulate our motivations. Overall, we hypothesize that a more realistic model will provide more accurate inferences on important epidemiological parameters such as the effective reproductive number. We also believe that our specific parameterization of the alarm function provides novel scientific insight on the relative importance of perceived susceptibility and perceived severity of the disease over time. We agree that unnecessary complexity is not useful, and believe our simulations are an important contribution that assesses which extensions are necessary and which are not. In our responses below we point to specific parts of the text that have been modified or added to address each of your comments.

1. Clarify why a fully stochastic population-averaged behavioral change model is needed as opposed to a partially stochastic or deterministic framework.

We have detailed the advantages of the fully stochastic Bayesian framework on Lines 26-28 and line 45.

2. Define a specific purpose for expanding the previous model to include hospitalizations and deaths compartments, while also accounting for undetected infections e.g. what answers do you hope to seek from an SIHRD structure with undetected infections that you couldn't get with the previous model structure?

We have detailed these motivations on Lines 62-70 (motivation for SIHRD structure) and lines 74-78 (motivation for undetected infections).

3. Clarify why the paper proposes a multivariable alarm with cases and deaths. Is this driven by a particular behavioral theory that individuals and policymakers respond to cases and deaths simultaneously or by empirical observations? Is there a specific hypothesis the authors are pursuing (e.g. including deaths in the alarm function will improve inference on $R_0(t)$) or is the motivation purely modeling improvements (more detailed alarm function to match more detailed model)?

Yes, this is driving by the health belief model which specifies risk perception is a function of both perceived susceptibility and perceived severity. We have clarified this on lines 53-61.

4. Clarify the expected benefits of the proposed extension compared to the previous model

We have clarified this on lines 68-70 and 76-78, and 81-85.

5. In introduction, paragraph 3, we need more information on why the SIR structure with behavioral alarm is simplifying. Depending on the model's purpose, these specifications can be adequate, such as for short-term prediction or exploratory analysis to understand the influence of behavior. Specify under what epidemiological or policy scenarios the assumptions are likely to be reasonable vs unrealistic, including the trade-offs relative to more complex structures.

We agree that the SIR model, although simpler than the SIHRD model, is extremely useful. In our refined introduction we motivate the more complex compartmental structure as a way to ensure the alarm function is only specified by endogenous inputs. We also have added some text on lines 51-52 about when the SIR structure can be useful and explicitly state that our goal is to understand whether the increased model complexity is actually needed or not (lines 83-85).

6. **Introduction, paragraph 4, lines 44-45.** An explanation of "fully stochastic" Bayesian model is needed here to help the reader understand why they include fewer compartments. Again emphasizing the tradeoffs that come with more complex model specifications.

In our revised introduction, we have removed the distinction between the fully stochastic and partially stochastic models as this didn't feel as relevant to our true motivations. We use the fully Bayesian framework as that is the foundational framework for the alarm-based behavioral change models that we consider and for the reasons outlined in response to your comment #1 in this section. **Previous implementations of these models used simpler compartmental structures as data augmented MCMC is needed when a compartment transition is not observed.** However, we have hospitalization and death data available so this is not a big concern for our extension and therefore we no longer think it is worth discussion in the introduction.

7. **Introduction, paragraph 4, lines 51-53.** The availability of hospitalization and death data certainly motivates the inclusion of these states in an epidemic model, but it is not clear why it motivates a fully stochastic model. The disease states can still be included in a partially stochastic or even fully deterministic model. Please provide specifics on why a fully stochastic modeling framework is needed to account for hospitalization and death states beyond just the availability of data.

As mentioned in our previous responses, the fully stochastic model is advantageous for several reasons, in particular the ability to propagate uncertainty from partially observed or unobserved disease states, which is a critical limitation when estimating important quantities such as the time-varying reproductive number that depend on the full, latent disease process.

8. **Introduction, paragraph 5.** The purpose of this paragraph is not entirely clear. You list several approaches that have been used to address undetected infections in a population model, then specify that a particular method used in individual models is infeasible for population models. It is not clear what purpose this serves. Revise the paragraph to clarify what gaps remain after the existing approaches used by population models and how your proposed modeling framework can help. Have a specific purpose for discussing methods used by individual level models (e.g. are they an improvement over population methods, are you trying to merge both approaches etc.) or leave them out.

I don't quite understand what we are saying here. [Not read paper yet, so it may become clearer]

We agree this paragraph was not very streamlined. We have removed the discussion of the individual level models and reframed the focus to the gap on incorporating undetected infections in endogenous behavioral change models (lines 74-78)

9. **Introduction closing paragraph.** Before the closing paragraph, you should have defined how the alarm used in the reference paper worked and its limitations, then state how using two data sources is an improvement on those limitations and what the data sources are.

We have added more specific details on the alarm function on lines 41-45. We state why we think two data sources will be an improvement on lines 53-61, and we are now specific about these data sources being cases and deaths throughout the introduction.

Major comment 2: Interpretation of SIHRD's better performance

10. The paper states that six models are fit, including an SIR model with a multivariable alarm based on cases and deaths, and also notes that deaths are not part of the SIR compartmental structure and cannot be simulated for posterior prediction. However, it is never made explicit how deaths actually enter the multivariable alarm for the SIR model. Without this, it is difficult to interpret why SIR performs poorly in high-death-importance scenarios. Clarify this implementation e.g. are deaths exogenous inputs to the multivariable alarm for the SIR case?

Yes, deaths are treated as an exogenous input for the SIR model with multivariable alarm. We have clarified this in both the introduction and on lines 177-180 of the methods section.

11. The paper attributes SIHRD's performance over SIR when deaths have more importance in the alarm function as evidence that it's more complex structure is beneficial. However, this result may reflect alignment between SIHRD's state variables and the alarm function. When the alarm depends heavily on deaths, models that include deaths as a state variable have more advantage. SIR does not model deaths and it is currently unclear how deaths are included in the multivariable alarm function. The interpretation of SIHRD's advantage needs to be revised in the manuscript to show that when a particular outcome strongly influences behavior, it needs to be included in the compartmental structure of the model for best results.

We have revised some of the interpretations of these results on lines 303-304 to clarify that we think the SIHRD advantage is because it can incorporate deaths as an endogenous input and the SIR cannot. We also revised the discussion lines 424-425 in a similar way.

12. This is a suggestion not requirement. In the simulation study, because the true alarm includes deaths, SIHRD models all the alarm inputs (cases & deaths) while SIR does not. This makes it difficult to assess whether SIHRD would still outperform SIR if behavioral responses depend only on signals common to both models. Consider an additional experiment where the true alarm depends on cases only. If SIHRD significantly outperforms SIR in this case, then the authors can confidently point to the richer complexity of SIHRD being an advantage.

Thank you for the suggestion. Our high case importance scenario is similar to this ($\alpha = 0.85$ vs the suggestion of $\alpha = 1$). Our simulations do indicate a slight benefit of the SIHRD model in this case in terms of the RMSE of R_0 , posterior predictive model fit, and estimation of the relative importance parameter α . Given these results, we don't feel that rerunning the simulations with $\alpha = 1$ would provide substantial new insight and believe that we can point to the results from the high case importance scenario to conclude that the SIHRD model still outperforms the SIR model.

Major comment 3: Alarm function ignores delays in behavioral response

Cases and deaths are not just correlated; deaths are a lagged outcome of cases. This makes it intrinsically difficult to separately identify their contributions to behavior, especially over short time horizons. In your framework, this challenge is compounded by the assumption that the alarm is an *instantaneous* function of current reported cases and deaths: alarm_t depends only on counts at time t , and β is modulated by $(1 - \text{alarm}_t)$. In reality, behavioral and policy responses typically accumulate information over time and respond with delays, not just to the current day's counts. Under your current specification, behavioral change that occurs with a delay after rising cases may be attributed to deaths simply because deaths occur later in time and may better align with the timing of the inferred change in $R_0(t)$.

Thank you for this comment as it pointed out to us that the methodology was not clearly explained. In our implementations, the alarm is based on a smoothed average of the past m days of reported cases and deaths and we used $m = 30$ which allows for a substantial lag. This was stated briefly in Section 2.1 when introducing the SIR model, and we have now added some text to further clarify that this is also used in the expanded alarm with case and deaths on lines 167-168. We have also added text to the data analysis section (line 349-350) to clarify that for these analyses we set $m = 30$ days.

Taken together, these features imply that estimated “death effects” in the alarm may partially capture delayed responses to previous cases rather than a distinct behavioral sensitivity to deaths per se. The Miami Wave 2 results highlight this concern: Miami experiences a substantial increase in cases and only a modest change in deaths, yet the alarm function attributes higher importance to deaths (a close to zero). As currently presented, it is difficult to conclude that deaths were the primary influencing factor of behavior without considering the possibility that the model is using deaths as a proxy for lagged responses to sustained high cases

We believe the clarification that the alarm is a function of 30-day smoothed case and death counts should address this concern. In addition, the lower WAIC for the model incorporating deaths in the alarm function for Miami Wave 2 suggests that deaths are a primary influencing factor since their incorporating significantly improves model performance.

13. In the Methods/Discussion, explicitly note that deaths are a lagged consequence of cases, and the alarm is an instantaneous function of current counts.

Explain that, in combination, these features may limit the ability to disentangle behavioral responses to cases versus deaths, and that estimated death effects may partly reflect delayed responses to cases rather than a purely independent response to deaths.

We have clarified in the methods (lines 167-168) that that alarm is not based on instantaneous function of current counts – rather it is based on a smoothed average over 30 days in both our simulations and data analysis.

14. Comment on the seemingly counterintuitive result that deaths are inferred to be the dominant driver of behavior in Miami Wave 2 despite the much larger increase in cases. Offer plausible explanations grounded in the model and data.

Thank you for noting that this could be seemingly counterintuitive to readers as that indicates that we did not explain this adequately. In fact, this result is not counterintuitive at all - if deaths are more important in influencing behavior and happen after cases, then it will take longer for a response to happen (e.g., for the alarm to increase), so case counts would be higher. This behavior was seen in our simulated data as well, where the observed case counts were much higher in the “High deaths importance” scenario (Figure 1).

We have provided further explanations about these results on lines 371-374.

15. If feasible, conduct a sensitivity check with more realistic alarm formulations that incorporates delays

Note that a more realistic behavioral model would incorporate delays—for example, by basing the alarm on moving averages or lagged inputs, see this study by [Zhang et al. \(2026\)](#), so that behavior at time t depends on recent history rather than only current-day counts. If feasible, consider a simple sensitivity analysis with a lagged or smoothed alarm to assess whether the main conclusions about the relative importance of cases versus deaths (particularly in Miami Wave 2) are robust. If this is beyond the scope of the current paper, suggest it as an important direction for future work and acknowledge that the current inferences could change under an alarm that considers delayed behavioral response.

Since the model already incorporates delays by basing the alarm on a moving average, we don't think this sensitivity analysis is necessary.

Major comment 4: Truncated epidemic history may affect behavioral response

The model is fit separately to Wave 1 and Wave 2. Cutting the time series at wave boundaries discards infection history that influences early deaths and behavior in the next wave. Fitting each wave independently, with its own initial conditions and parameters, underplays behavioral “memory” from Wave 1 into Wave 2 through evolving risk perception, fatigue, etc. As such, this design does not fully assess how the alarm behaves over a continuous, multi-wave epidemic.

It is true that modeling individual waves separately limits our ability to assess long-term behavioral patterns. We made this decision for three reasons, first, we hypothesized that the relative importance of cases and deaths would change over time and the only way to assess this in our current framework is to fit separate multivariable alarm models to each wave. Secondly, modeling COVID-19 over longer durations of time would necessitate further extensions to the model to allow for detection rates to change over time and for baseline transmissibility to change over time with the introduction of new variants. These are important extensions which are beyond the scope of our current work. We believe that the more specific framing of the introduction will help specify the scope of our work. We have also added a new paragraph to the discussion specifically about modeling multiple waves simultaneously and the relative tradeoffs of that approach versus what we did.

16. Briefly discuss in the main text how fitting each wave separately might affect identification of the alarm parameters and potentially overstate how well the alarm can explain long-term behavior when epidemic history is truncated.

This is addressed in our new discussion paragraph on Lines 470-479.

17. Line 290. Summarize (in the main text) what you observed when fitting different time windows: were the identified alarm parameters and $R_0(t)$ trajectories broadly consistent, or did they vary across fitting windows?

Commented [CW3]: I'm actually not clear what this is referring to on Line 290...

The estimated alarm parameters and $R_0(t)$ trajectories from the two time periods we modeled are presented in Figure 7 and discussed on Lines 397-411.

Minor comments

- In line 27, "behavioral parameters using motivated by a real observed epidemic". Delete using OR motivated by. **fixed**
- In introduction, lines 30-33, check if this recent review can provide additional helpful information: Hamilton A, Haghpanah F, Tulchinsky A, Kipshidze N, Poleon S, Lin G, Du H, Gardner L, Klein E. Incorporating endogenous human behavior in models of COVID-19 transmission: A systematic scoping review. *Dialogues Health*. 2024 May 7;4:100179. doi: 10.1016/j.dialog.2024.100179. PMID: 38813579; PMCID: PMC11134564. **Thank you. This is a useful reference which we have cited in the introduction on Lines 35-38.**
- The model is formulated as a discrete-time, fully stochastic chain-binomial/multinomial process. Brief notes on how this approach compares to continuous-time ODE formulations will be helpful to clarify the advantages of your approach to readers.
- Methods paragraph 2.** Briefly specify how the choice to use a power alarm was made in the previous paper e.g. based on model selection criteria, identifiability, prediction performance, interpretability etc **Previous work has used WAIC to select the alarm specification among several other monotonic functions, but the best alarm function varied depending on the observed data. We chose the power alarm for this work because the more parsimonious representation was helpful for incorporating multiple data sources as described in the supplementary material Section 6. We've added a brief comment about this in the methods section which refers readers to the discussion in the supplement for more details.**
- Methods line 137.** Insert comma after t **fixed**
- Methods line 149.** Provide references showing that vague priors for beta and k are a standard choice in the literature or include a rationale for the choice. **fixed**
- Methods 2.4 Paragraph 1.** It would be helpful to see the type of infectiousness curves that the prior for ν generates so readers can better understand its epidemiological

Except that....
I'm not sure there is a particular advantage to modelling in DT or CT. ODE assumes S, E, I, R , etc are real which means you can have parts of individuals infected. The main advantage is the stochastic nature of our model. I'd say...

Commented [CW4]: Rob? I know you have done continuous time modeling. I don't really know how this model relates to the continuous time ODE so not sure what to say here.

as uncertainty qualification of all incorporating levels; prior knowledge; better for small pop's; more difficult to capture BC feedback in ODE SIR

discrete time also more natural for the data collection process (with ODE need an additional observation model)

implications. Did you perform any sensitivity analysis around this choice? **We have added a supplementary figure illustrating this choice. We did not perform any sensitivity analysis for this choice as we feel our choice is defensible and the specific shape of this curve ultimately has little impact on the estimated reproductive number as illustrated by Ward et al 2022 in their original implementation.**

Ward, C., Brown, G. D., & Oleson, J. J. (2023). Incorporating infectious duration-dependent transmission into Bayesian epidemic models. *Biometrical journal*, 65(3), 2100401.

8. **Methods line 170.** More clarification is needed on the prior choice for detection probability. **We have added more specific details about the specification of this prior on Lines 221-223**
9. **Methods 2.5.** This entire section needs revision. Currently you provide a general description of approaches for model selection instead of a clear description of what you did in the paper. Please revise to specify the actual methods for assessing model fit used for the SIHRD model (which is the focus of this paper). **Posterior predictive assessment and WAIC are described here as they are used in both the simulation study and data analysis to assess and compare model fit. We have revised the text to sound less like a general description and more specific to our implementations.**
10. **Discussion line 356.** Specify the data sources, cases and deaths. **done**
11. **Discussion last two paragraphs.** Two paragraphs describing alternative specifications of the alarm function is not needed here. This detail can be pushed to the method or supplementary section. A simple statement acknowledging limitations of the current choice and possibilities for improvement is sufficient for the discussion, especially when there are more important things to be discussed. I suggest removing these paragraphs. **We have moved these to the supplement.**